

1. Why is Feature Selection Important?

- Removes unnecessary columns
- Reduces complexity
- Improves prediction accuracy
- Speeds up training

2. Train-Test Split

After feature selection:

```
#Feature Selection
X = df[['bedrooms',
        'bathrooms',
        'square_feet',
        'lot_size_sqft',
        'year_built',
        'garage_spaces',
        'distance_to_city_center_km',
        'crime_rate_index',
        'nearby_school_rating',
        'days_on_market']]

y = df['house_price']

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42
)
```

Explanation:

"I split the dataset into 80% training data and 20% testing data. The training data teaches the model patterns, while the testing data evaluates how well the model performs on unseen data."

3. Model Creation:

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(X_train, y_train)
```

Explanation:

"Model creation is the process of selecting a machine learning algorithm and training it using the training dataset so it can learn the relationship between input features and the target."

For house price prediction, the most common algorithm is **Linear Regression** because the target (price) is a continuous value.

4. Prediction:

```
y_pred = model.predict(X_test)
```

"After training, I used the testing dataset to predict house prices. These predictions are compared with the actual prices to evaluate model performance."

5. Model Evaluation:

```
from sklearn.metrics import r2_score, mean_absolute_error  
  
print("R2 Score:", r2_score(y_test, y_pred))  
print("MAE:", mean_absolute_error(y_test, y_pred))
```

Explanation:

- **MAE (Mean Absolute Error):** Average prediction error. Lower is better.
- **R² Score:** Measures how well the model explains the variation in house prices. Closer to 1 indicates a better fit.